Assignment 8

Caden O'Leary, Elmoataz Abdellah, Matthew Goldsberry, Seunghun Lee, Kristen Griessman

Task List

1. Design the static memory buffer allocation scheme for the pyld_data and imgr_data queues on the **Main Computer**. (Matthew Goldsberry)

2. Integrate an On-Orbit Flight Software Reprogrammability if Main Computer's main mission operations ever corrupts on **Main Computer** (Seunghun Lee).

3. Develop the core active_task chlorophyll index computation algorithm for the **Data Processor** module. (Matthew Goldsberry)

4. Document the API contract and error handling responses for passing health_status from the **Data Processor to the Main Computer**. (Matthew Goldsberry)

5. Validate the inter-component CAN bus communication latency between the **Data Processor** and the **Main Computer**. (Matthew Goldsberry)

6. Research the I2C mutex locks needed for the ADCS Computer to prevent bus collisions during **attitude sensor** reads. (Elmoataz Abdellah)

7. Specify the exact 12-byte packet structure for sending adcs_data (including pos_x, pos_y, and pos_z) from the **ADCS Computer** to the **Main Computer**. (Elmoataz Abdellah)

8. Simulate a setting for the development environment to run isolated unit tests on **Main Computer's** communication with other subsystem components (Elmoataz Abdellah).

9. Develop the F Prime Svc::TlmChan configuration to channelize the vel_x, vel_y, and vel_z attributes within the **ADCS Computer**. (Elmoataz Abdellah)

10. Refine the faults handling logic for the **ADCS Computer** to correctly flag sensor dropouts to the **Main Computer** without triggering a full system reset. (Elmoataz Abdellah)

11. Test the command ingestion pipeline routing goal_pos_y and goal_pos_z targets from the **Main Computer** to the **ADCS Computer**. (Elmoataz Abdellah)

12. Investigate the SPI polling rate required for the **Main Computer** to accurately read the **Power Computer's** charge_percentage and voltage attributes. (Caden O'Leary)

13. Design the time_to_empty estimation algorithm to run iteratively on the **Power Computer**. (Caden O'Leary)

14. Document the state transitions and recovery steps for the faults attribute within the **Power Computer** module. (Caden O'Leary)

15. Develop the hardware watchdog petting sequence between the **Power Computer** and the **Main Computer's** health_status buffer. (Caden O'Leary)

16. Validate the power subsystem's fault isolation logic, ensuring the **Power Computer** properly updates its time_stamp during transient voltage spikes. (Caden O'Leary)

17. Research the maximum SPI bus throughput for writing truth_imgr_data from the **Imager Adapter** to the **Main Computer**. (Seunghun Lee)

18. Specify the byte alignment and frame dimensions for the multi_imgr_data frames outputted by the **Imager Adapter**. (Seunghun Lee)

19. Design the circular buffer mechanism to accurately track packets_processed inside the **Data Storage** module. (Seunghun Lee)

20. Develop the file chunking and active_mode flagging logic for the **Data Storage** component to prepare files for downlink. (Seunghun Lee)

21. Test the temp and health_status reporting mechanisms of the **Data Storage** unit under simulated high-I/O read/write loads. (Seunghun Lee)

22. Obtain the required SQLite ORM libraries and setup the ground station database to receive pwr_data telemetry from the **Main Computer**. (Elmoataz Abdellah)

23. Specify the JSON schemas for the Ground Station API to exactly match the adcs_data and imgr_data structures buffered on the **Main Computer**. (Kristen Griessman)

24. Ensure security through Encryption, Decryption, and Authentication of packets by the **Main Computer** to and from the **Ground Station**. (Elmoatatz Abdellah)

25. Refine the **Ground Station** packet decoding logic to handle fragmented pyld_data transmissions stored in the **Data Storage** module. (Caden O'leary)

26. Validate the end-to-end data pipeline from the **Main Computer's** buffer to the **Ground Station's** database during a simulated pass execution. (Kristen Griessman)

27. Test the inter-process communication serialization between the **Main Computer** and **Data Processor** modules under maximum telemetry load. (Matthew Goldsberry)

Milestones List

SRR (Complete)

Explanation: System requirements review. HABSat-1 has undergone a system requirement review from a board of members in the aerospace industry to receive feedback.
Date: 4/23/26
Verifier: UC CubeCats and industry experts

Assignment 1 (Complete)

Explanation: Professional biography
Date: 9/2/26
Verifier: Team and graders

Assignment 2 (Complete)

Explanation: Team formation and project description
Date: 9/9/26
Verifier: Team and graders

Assignment 3 (Complete)

Explanation: Advisor selection and design constraints essay
Date: 9/18/26
Verifier: Team and graders

Assignment 4 (Complete)

Explanation: User stories and use cases
Date: 9/28/26
Verifier: Team and graders

Assignment 5 (Complete)

Explanation: High-level system design (Design D0)
Date: 10/2/26
Verifier: Team and graders

Assignment 6 (Complete)

Explanation: Detailed system design, part 1 (Design D1)
Date: 10/8/26
Verifier: Team and graders

Assignment 7 (Complete)

Explanation: Detailed system design, part 2 (Design D2) and peer review notes
Date: 10/14/26
Verifier: Team and graders

Assignment 8

Explanation: Task list, milestones, timeline, and effort matrix
Date: 10/28/26
Verifier: Team and graders

Assignment 9

Explanation: Requirements document and technical specification document
Date: 10/30/26
Verifier: Team and graders

Assignment 10

Explanation: Test and validation plan draft
Date: 11/4/26
Verifier: Team and graders

Assignment 11

Explanation: Ethics and responsible-use statement

Date: 11/11/26
Verifier: Team and graders

Assignment 12

Explanation: Presentation draft and documented advisor feedback
Date: 11/18/26
Verifier: Team and graders

Assignment 13

Explanation: Team oral presentation, recorded video, and peer reviews
Date: 11/18/26
Verifier: Team and graders

Assignment 14

Explanation: Design report, design notebook, and team and self assessment
Date: 12/2/26
Verifier: Team and graders

Assignment 15

Explanation: Project advisor assessments
Date: 12/8/26
Verifier: Team and graders

PDR

Explanation: Preliminary design review. HABSat-1 will undergo a preliminary design review from a board of members in the aerospace industry
Date: (TBD) Spring 2027
Verifier: UC CubeCats and industry experts

Power Interaction Prototype Software

Explanation: Finished prototype software with test cases for interactions between the main computer and the power subsystem components.
Date: 1/29/27
Verifier: Team and UC CubeCats

Comms Interaction Prototype Software

Explanation: Finished prototype software with test cases for interactions between the main computer and the comms subsystem components.
Date: 2/8/27
Verifier: Team and UC CubeCats

Payload Interaction Prototype Software

Explanation: Finished prototype software with test cases for interactions between the main computer and the payload subsystem components.
Date: 3/1/27
Verifier: Team and UC CubeCats

ACDS Interaction Prototype Software

Explanation: Finished prototype software with test cases for interactions between the main computer and the ACDS subsystem components.
Date: 3/15/27
Verifier: Team and UC CubeCats

Senior Design Expo

Explanation: Present finished project
Date: 4/13/27
Verifier: Team, UC CubeCats, students, professor

Timeline

Task ID	Statement	Owner	Start Date	Completion Date
001	Design the static memory buffer allocation scheme for the pyld_data and imgr_data queues on the Main Computer.	Matthew Goldsberry	1/13/27	1/16/27
002	Integrate an On-Orbit Flight Software Reprogrammability if Main Computer's main mission operations ever corrupts on Main Computer.	Seunghun Lee	1/13/27	1/16/27
003	Develop the core active_task chlorophyll index computation algorithm for the Data Processor module.	Matthew Goldsberry	1/16/27	1/20/27
004	Document the API contract and error handling responses for passing health_status from the Data Processor to the Main Computer.	Matthew Goldsberry	1/20/27	1/24/27
005	Validate the inter-component CAN bus communication latency between the Data Processor and the Main Computer.	Matthew Goldsberry	1/24/27	1/27/27
006	Research the I2C mutex locks needed for the ADCS Computer to prevent bus collisions during attitude sensor reads.	Elmoataz Abdellah	1/27/27	1/31/27
007	Specify the exact 12-byte packet structure for sending adcs_data (including pos_x, pos_y, and pos_z) from the ADCS Computer to the Main Computer.	Elmoataz Abdellah	1/31/27	2/4/27
008	Develop the F Prime Svc::TlmChan configuration to channelize the vel_x, vel_y, and vel_z attributes within the ADCS Computer.	Elmoataz Abdellah	2/4/27	2/8/27
009	Simulate a setting for the development environment to run isolated unit tests on Main Computer's communication with other subsystem components.	Elmoataz Abdellah	2/8/27	2/11/27
010	Refine the faults handling logic for the ADCS Computer to correctly flag sensor dropouts to the Main Computer without triggering a full system reset.	Elmoataz Abdellah	2/8/27	2/11/27

Task ID	Statement	Owner	Start Date	Completion Date
011	Test the command ingestion pipeline routing goal_pos_y and goal_pos_z targets from the Main Computer to the ADCS Computer.	Elmoataz Abdellah	2/11/27	2/15/27
012	Investigate the SPI polling rate required for the Main Computer to accurately read the Power Computer's charge_percentage and voltage attributes.	Caden O'Leary	2/15/27	2/19/27
013	Design the time_to_empty estimation algorithm to run iteratively on the Power Computer.	Caden O'Leary	2/19/27	2/22/27
014	Document the state transitions and recovery steps for the faults attribute within the Power Computer module.	Caden O'Leary	2/22/27	2/26/27
015	Develop the hardware watchdog petting sequence between the Power Computer and the Main Computer's health_status buffer.	Caden O'Leary	2/26/27	3/2/27
016	Validate the power subsystem's fault isolation logic, ensuring the Power Computer properly updates its time_stamp during transient voltage spikes.	Caden O'Leary	3/2/27	3/6/27
017	Research the maximum SPI bus throughput for writing truth_imgr_data from the Imager Adapter to the Main Computer.	Seunghun Lee	3/6/27	3/10/27
018	Specify the byte alignment and frame dimensions for the multi_imgr_data frames outputted by the Imager Adapter.	Seunghun Lee	3/10/27	3/13/27
019	Design the circular buffer mechanism to accurately track packets_processed inside the Data Storage module.	Seunghun Lee	3/13/27	3/17/27
020	Develop the file chunking and active_mode flagging logic for the Data Storage component to prepare files for downlink.	Seunghun Lee	3/17/27	3/21/27
021	Test the temp and health_status reporting mechanisms of the Data Storage unit under simulated high-I/O read/write loads.	Seunghun Lee	3/21/27	3/25/27
022	Obtain the required SQLite ORM libraries and setup the ground station database to receive pwr_data telemetry from the Main Computer.	Elmoataz Abdellah	3/25/27	3/28/27
023	Specify the JSON schemas for the Ground Station API to exactly match the adcs_data and imgr_data structures buffered on the Main Computer.	Kristen Griessman	3/28/27	4/1/27

Task ID	Statement	Owner	Start Date	Completion Date
024	Ensure security through Encryption, Decryption, and Authentication of packets by the Main Computer to and from the Ground Station.	Elmoataz Abdellah	4/1/27	4/5/27
025	Refine the Ground Station packet decoding logic to handle fragmented pyld_data transmissions stored in the Data Storage module.	Caden O'Leary	4/5/27	4/9/27
026	Validate the end-to-end data pipeline from the Main Computer's buffer to the Ground Station's database during a simulated pass execution.	Kristen Griessman	4/9/27	4/11/27
027	Test the inter-process communication serialization between the Main Computer and Data Processor modules under maximum telemetry load.	Matthew Goldsberry	4/11/27	4/13/27

Effort Matrix

Task	Elmoataz Abdellah	Matthew Goldsberry	Caden O'leary	Seunghun Lee	Kristen Griessman	Total (hrs)
001: Static Memory Buffer	2	**10 (owner)**	2	1	1	16
002: On-Orbit Software Reprogrammability	2	3	2	**11 (owner)**	2	20
003: Active Task Chlorophyll Alg	1	**12 (owner)**	1	4	2	20
004: API Contract & Health Status	3	**10 (owner)**	2	2	3	20
005: CAN Bus Latency Validation	3	**8 (owner)**	1	2	1	15
006: I2C Mutex Locks	**10 (owner)**	2	2	1	1	16
007: 12-Byte Packet Structure	**9 (owner)**	3	1	1	2	16
008: F Prime Svc::TlmChan Config	**12 (owner)**	2	2	2	2	20
009: Simulate Dev Environment	**11 (owner)**	3	2	2	2	20
010: ADCS Fault Handling Logic	**10 (owner)**	3	2	1	2	18
011: Command Ingestion Pipeline	**11 (owner)**	4	1	2	2	20
012: Power SPI Polling Rate	2	2	**10 (owner)**	1	1	16

Task	Elmoataz Abdellah	Matthew Goldsberry	Caden O'leary	Seunghun Lee	Kristen Griessman	Total (hrs)
013: Time to Empty Algorithm	2	2	**12 (owner)**	2	2	20
014: Power Fault Transitions	2	3	**10 (owner)**	1	2	18
015: Watchdog Petting Sequence	3	4	**11 (owner)**	1	1	20
016: Power Fault Isolation Logic	2	2	**12 (owner)**	2	2	20
017: Imager SPI Throughput	2	3	1	**9 (owner)**	1	16
018: Imager Byte Alignment	1	2	1	**9 (owner)**	2	15
019: Storage Circular Buffer	2	3	1	**12 (owner)**	2	20
020: File Chunking & Downlink	2	3	2	**11 (owner)**	2	20
021: Data Storage High-I/O Load	2	3	2	**11 (owner)**	2	20
022: Ground Station SQLite ORM	**10 (owner)**	2	1	1	2	16
023: API JSON Schemas	2	2	1	2	**13 (owner)**	20
024: React Front-End Dashboard	**12 (owner)**	2	2	2	2	20
025: Ground Station Packet Decoding	2	2	**11 (owner)**	2	3	20
026: End-to-End Pipeline Validation	2	2	2	2	**12 (owner)**	20
027: IPC Serialization Load Test	2	**10 (owner)**	1	1	1	15
Total Hours	**128**	**108**	**92**	**83**	**50**	**461**

Workload Balance Note: The higher total estimated hours for Elmoataz Abdellah (128 hrs) and Matthew Goldsberry (108 hrs) reflect their primary ownership of core software architecture, embedded controls, and ground station integration tasks. Kristen Griessman (50 hrs) and Seunghun Lee (83 hrs) will assist in peer review, system test harness development, and documentation during peak development phases to ensure overall team workload is balanced dynamically across the project lifecycle.

Work breakdown structure

Phase 1: Core Processing & Main Computer

1.1 Memory Allocation & Core Algorithms

- Task 001: Design the static memory buffer allocation scheme for pyld_data and imgr_data queues on the Main Computer
- Task 002: Integrate an On-Orbit Flight Software Reprogrammability if Main Computer's main mission operations ever corrupts on Main Computer
- Task 003: Develop the core active_task chlorophyll index computation algorithm for the Data Processor module

1.2 Inter-Component Communication & Diagnostics

- Task 004: Document the API contract and error handling responses for passing health_status from Data Processor to Main Computer
- Task 005: Validate the inter-component CAN bus communication latency between Data Processor and Main Computer
- Task 027: Test the inter-process communication serialization between Main Computer and Data Processor modules under maximum telemetry load

Phase 2: ADCS & Power Subsystems

2.1 Attitude Determination & Control System (ADCS)

- Task 006: Research I2C mutex locks needed for ADCS Computer to prevent bus collisions during attitude sensor reads
- Task 007: Specify the exact 12-byte packet structure for sending adcs_data (including pos_x, pos_y, pos_z) from ADCS Computer to Main Computer
- Task 008: Develop the F Prime Svc::TlmChan configuration to channelize vel_x, vel_y, and vel_z attributes within ADCS Computer
- Task 009: Simulate a setting for the development environment to run isolated unit tests on Main Computer's communication with other subsystem components

- Task 010: Refine faults handling logic for ADCS Computer to correctly flag sensor dropouts to Main Computer without triggering full system reset
- Task 011: Test command ingestion pipeline routing goal_pos_y and goal_pos_z targets from Main Computer to ADCS Computer

2.2 Power Subsystem & Watchdog Controls

- Task 012: Investigate SPI polling rate required for Main Computer to accurately read Power Computer's charge_percentage and voltage attributes
- Task 013: Design time_to_empty estimation algorithm to run iteratively on Power Computer
- Task 014: Document state transitions and recovery steps for faults attribute within Power Computer module
- Task 015: Develop hardware watchdog petting sequence between Power Computer and Main Computer's health_status buffer
- Task 016: Validate power subsystem's fault isolation logic, ensuring Power Computer properly updates time_stamp during transient voltage spikes

Phase 3: Imaging Subsystem & Data Storage

3.1 Imager Adapter Interfaces

- Task 017: Research maximum SPI bus throughput for writing truth_imgr_data from Imager Adapter to Main Computer
- Task 018: Specify byte alignment and frame dimensions for multi_imgr_data frames outputted by Imager Adapter

3.2 Data Storage & Downlink Management

- Task 019: Design circular buffer mechanism to accurately track packets_processed inside Data Storage module
- Task 020: Develop file chunking and active_mode flagging logic for Data Storage component to prepare files for downlink
- Task 021: Test temp and health_status reporting mechanisms of Data Storage unit under simulated high-I/O read/write loads

Phase 4: Ground Station & Pipeline Integration

4.1 Ground Station Infrastructure & Front-End

- Task 022: Obtain required SQLite ORM libraries and setup ground station database to receive pwr_data telemetry from Main Computer

- Task 023: Specify JSON schemas for Ground Station API to exactly match adcs_data and imgr_data structures buffered on Main Computer
- Task 024: Develop React front-end dashboard to actively poll and display health_status of Main Computer and Power Computer
- Task 025: Refine Ground Station packet decoding logic to handle fragmented pyld_data transmissions stored in Data Storage module

4.2 End-to-End System Validation

Task 026: Validate end-to-end data pipeline from Main Computer's buffer to Ground Station's database during simulated pass execution

Risk Register

Unmitigated Risk Matrix

Occurance	1	2	3	4	5
5				5	
4			7, 9, 20	6	1
3			8, 13, 16, 17, 18	4, 10, 12, 14, 15, 19, 21, 24, 27, 28, 29, 30, 32	2, 3, 25
2					11, 22, 23, 26, 31
1					

Consequence / Severity

Mitigated Risk Matrix

Occurance	1	2	3	4	5
5					
4					
3					
2		6, 20, 21, 28, 29, 30	1, 5		
1		4, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19, 24, 27, 31, 32	2, 3, 11, 22, 23, 25, 26		

Consequence / Severity

ID	Design Item	Failure	Effect(s)	Severity	Cause(s)	Occourance	Current Control(s)	Critical Number	Risk Priority	Category	Strategy	Trigger	Owner
1	Main Computer Flight Software & RAM [L2-SYS-059, L3-OBC-159, L4-OBC-251]	Radiation-induced Single-Event Upset (SEU) bit flips corrupt active flight software execution or dynamic RAM memory structures	Unhandled software exception, task crash, or processor hang halts Command & Data Handling (C&H) and mission sequencing	5	Ionizing radiation in 350–550 km LEO affecting commercial Cortex-M4 memory cells	4	EnduroSat board shielding (L3-OBC-163), internal processor watchdog (L4-OBC-250), and FreeRTOS task memory protection	24	-	Technical	Reduce	Processor watchdog reset or unhandled RTOS HardFault exception	Seunghun Lee & Matthew Goldsberry
2	Main Computer NVM & Boot Flash [L3-OBC-166, L4-OBC-251]	Single-Event Latch-up (SEL) or flash memory wear corrupts primary flight OS image in boot flash	Undockable primary flight computer image halts spacecraft boot cycle and mission operations	5	Heavy-ion strike or permanent latch-up in non-rad-hardened NOR boot flash array	3	File CRC verification (L4-OBC-224), golden boot image fallback (L4-OBC-251), and hardware shielding	21	-	Technical	Reduce	Bootloader CRC check failure on configuration/flight image or 3 consecutive failed boots (L4-OBC-251)	Seunghun Lee & Matthew Goldsberry
3	Internal CAN Bus Network - Transceivers [L3-OBC-160, L4-OBC-252]	Shared CAN bus physical lockup or dominant-bit latch caused by a faulted peripheral transceiver	Blocks deterministic 12-byte command and telemetry packets across ADCS, EPS, and Payloads, halting health polling	5	Multi-drop CAN topology allows a single failed peripheral node transceiver to pull the differential bus dominant	3	CAN 2.0B hardware frame check and EPS autonomous peripheral power isolation	21	-	Technical	Reduce	CAN controller Tx error counter exceeds 255 or bus dominant state persists >500 ms	Matthew Goldsberry
4	Internal CAN Bus Network - Controller & Arbitration [L4-OBC-252, L4-OBC-263]	Transient EMI noise spikes induce repeated transmission frame errors, forcing CAN controller into bus-off state	Halts subsystem telemetry collection and inter-module command distribution until controller reset	4	High-power switching transients from EPS or RF transmitters coupling into unshielded bus harness	3	CAN 2.0B hardware CRC, bit stuffing, frame check, and up to 3 automatic transmission retries	19	-	Technical	Reduce	CAN controller enters bus-off state across 3 consecutive local reset attempts (L4-OBC-252, L4-OBC-263)	Matthew Goldsberry
5	Timeline Critical Path - Core Processing	Schedule slip in Phase 1 Core Processing tasks cascades across subsequent tasks	Misses subsystem interaction prototype deadlines (1/29 Power, 2/8 Comms, 3/1 Payload, 3/15 ADCS)	4	Tight 3–4 day sequential task windows with zero slack between serial dependencies and student exam conflicts	5	Weekly Wednesday team meetings, Tuesday/Sunday lab sessions, 24-hour Discord policy, and WBS Gantt tracking	23	-	Team	Reduce	Any Phase 1 Core Processing task slips >48 hours past its scheduled deadline	Matthew Goldsberry
6	Timeline Critical Path - NASA F' Integration	NASA F' framework autocoding learning curve and FreeRTOS integration complexity delay serial Table 1 tasks	Jeopardizes 4/13/27 Senior Design Expo demo and HABSat-1 PDR readiness	4	Steep learning curve with NASA F' C++ autocoder, XML topology dictionaries, and OSAL abstractions	4	NASA F' GDS tutorials, unit test suite, and peer-programming sprint sessions	22	-	Team	Reduce	NASA F' component integration or topology build blocked >72 hours on critical path	Matthew Goldsberry
7	Club Onboarding & Software Toolchain [L0-EDU-005]	Steep onboarding barrier for C++, FreeRTOS, and NASA F' environment setup deters club contributors	Low CubeCats member retention and loss of continuity	3	Complex embedded cross-compilation toolchain on heterogeneous student operating systems	4	Shared GitHub repository and Google Drive design notebook documentation	18	-	Team	Avoid	New member dev environment setup takes >2 weeks to compile/run a first F' build	Caden O'Leary
8	Lab Testbench Wiring & Hardware Documentation	Undocumented breadboard/UART jumper wiring changes between lab sessions cause accidental wiring faults	Accidental hardware damage, damaged MCU GPIO pins, and bench testing downtime	3	Uncommunicated physical breadboard and power supply wiring modifications between Tuesday and Sunday labs	3	Tuesday/Sunday in-person lab sessions and shared hardware schematic notebook	15	-	Team	Avoid	Lab testbench wiring modified without updating pinout schematic or wiring discrepancy found	Caden O'Leary
9	Dev Board Breadboard Signal Integrity [L4-OBC-256]	Breadboard jumper wire signal degradation, parasitic capacitance, or loose connections cause bus errors	Intermittent bus communication failures during engineering model hardware-in-the-loop test runs	3	High-speed SPI/UART signals routed through long jumper wires and standard breadboard tie-points	4	Breadboard pinout cross-reference mapping and logic analyzer signal verification	18	-	Technical	Avoid	Breadboard bus signal integrity failure or peripheral pin assignment mismatch with flight schematic	Elmoataz Abdellah
10	Dev MCU vs Flight EnduroSat Board Fidelity	Clocking, memory, or peripheral register timing discrepancies between dev MCU and EnduroSat flight board delay bug detection	Integration defects discovered late during final flight board acceptance testing	4	EnduroSat flight boards quarantined from routine bench testing; simulation relies on development board	3	NASA F' framework with Ground Data System (GDS) unit test suite	19	-	Technical	Avoid	Off-target unit test coverage drops below 80% (L4-OBC-256) or flight board acceptance anomaly	Elmoataz Abdellah
11	Main Computer Hardware Failure & EPS Backup Radio Link [L2-SYS-062, L3-POW-187, L4-OBC-258]	Catastrophic hardware failure or unrecoverable lockup of Main Computer severs primary command and telemetry	Total loss of primary spacecraft commandability unless secondary subsystem failover operates	5	Launch vibration, thermal stress, or permanent latch-up on primary flight computer PCB	2	Space-qualified EnduroSat OBC Type I board and EPS critical channel power recovery (L3-POW-187)	17	-	Technical	Transfer	Main Computer fails to transmit periodic heartbeat to EPS within L4-POW-174 timeout after EPS power-cycle	Caden O'Leary & Seunghun Lee
12	Watchdog Architecture - 3-WDT Majority Voting [L4-OBC-247, L4-OBC-259]	Desynchronization or timing drift among the 3 concurrent WDTs triggers false-positive Safe Mode trip	Unintended flight computer reboots mid-pass and disruption of active payload data collection	4	Timing drift between internal hardware WDT, external windowed WDT, and F' software supervisor	3	FreeRTOS preemptive priority scheduling (L4-OBC-247) and wrapped NASA F' Svc::Health component	19	-	Technical	Reduce	1 of 3 WDTs disagrees with the other 2 (2-of-3 majority mismatch)	Matthew Goldsberry & Caden O'Leary
13	Watchdog Architecture - Dynamic UART Lines [L4-OBC-253]	Unmonitored communication hang or serial lockup on dynamically toggled UART interface lines	Hung serial communication threads with UHF Radio or Secondary Payload stalling command pass	3	Missing watchdog supervisor and timeout handling on dynamically multiplexed UART lines	3	NASA F' Svc::Health component and software timeout counters on serial read routines	15	-	Technical	Reduce	Dynamic UART state transition timeout or unserviced serial thread blocked >500 ms	Matthew Goldsberry & Caden O'Leary
14	Memory Management - Runtime SRAM Exhaustion [L3-OBC-144, L4-OBC-248–249]	Runtime SRAM memory utilization exceeds 95% capacity under concurrent workload	RTOS stack overflow, unallocated telemetry packets dropped, or unhandled software crash	4	Constrained SRAM on Main Computer chip handling concurrent frame buffering and communication queues	3	Zero dynamic heap allocation policy in flight software (L4-OBC-248)	19	-	Technical	Reduce	Runtime RAM utilization exceeds 95% threshold alert (L4-OBC-249)	Elmoataz Abdellah & Seunghun Lee
15	Flash Mass Storage - Storage Saturation & Wear [L2-SYS-042, L4-OBC-229–230]	Flash mass storage saturation (>99%) or bad NAND flash blocks corrupt stored Lake Erie imagery	Buffer overflow, dropped multispectral science frames, or loss of historical fault/telemetry logs	4	Unretrieved imagery images until Ground Station handshake; concurrent SPI writes, and flash block wear	3	Ground Station verified purge handshake (L2-MIS-084, L4-OBC-225) and flash wear-leveling driver	19	-	Technical	Reduce	Flash storage utilization >85% warning (L4-OBC-229), >99% capacity (L4-OBC-230), or SPI flash CRC error	Elmoataz Abdellah & Seunghun Lee
16	Storage Bus - SPI Bus Contention [L3-OBC-148, L4-OBC-263]	SPI bus contention between high-rate Imager writes and Telemetry logging threads	Corrupted SPI flash write packets, telemetry log gap, or imager frame buffer overflow	3	Multiple tasks accessing shared hardware SPI peripheral without strict serial mutex locks	3	Hardware mutex locks and task priority separation on shared SPI bus	15	-	Technical	Reduce	SPI transaction lock timeout or bus collision flag asserted during active pass	Elmoataz Abdellah & Seunghun Lee
17	Secondary Data Processor - Math Guard [L4-OBC-246]	Data Processor arithmetic divide-by-zero or overflow during Chlorophyll Index calculation	Corrupted or delayed HAB science data products halting image compression pipeline	3	Anomalous sensor pixel values or saturated raw inputs during spectral index division	3	Offloading heavy image processing from Main Computer to Data Processor	15	-	Technical	Reduce	Arithmetic divide-by-zero or numeric overflow exception trap (L4-OBC-246)	Matthew Goldsberry & Seunghun Lee
18	Secondary Data Processor - Execution Timeout [L4-OBC-261–262]	Data Processor hangs or exceeds execution timeout budget during Lake Erie image processing	Excessive peak power draw (L3-OBC-158) and processed HAB products not ready for scheduled downlink	3	DSP algorithm hang or corrupted UART frames from Imager Adapter	3	Up to 3 UART frame block retries (L4-OBC-245) and hardware execution watchdog	15	-	Technical	Reduce	Data Processor execution timer expiry (L4-OBC-262) or task watchdog timeout	Matthew Goldsberry & Seunghun Lee
19	High-Rate Downlink - S-Band Bus Contention [L3-OBC-187, L4-OBC-200]	Main Computer and Data Processor contend for S-Band radio access causing buffer underrun/stall	DMA buffer underrun/stall during high-rate S-Band image downlink drops RF carrier mid-pass	4	Shared S-Band interface with asynchronous DMA flow control and competing data sources	3	Dedicated transmitter FIFO buffering and scheduled radio access windows	19	-	Technical	Reduce	S-Band transmitter buffer underrun or overflow flag asserted (L4-OBC-200)	Matthew Goldsberry & Seunghun Lee
20	Orbit Synchronization - GPS & Epoch Clock Sync [L2-SYS-038, L4-OBC-217]	GPS lock or 1PPS signal dropout during orbital eclipse causes epoch clock timing drift	Timing uncertainty degrades georeferencing metadata and orbit propagation accuracy	3	CubeSat occultation during eclipse and low GPS antenna gain in off-nominal attitudes	4	ADCS 3D-minute orbit propagation schedule packets (0x43) and Ground Station pre-pass TLE filtering	18	-	Technical	Reduce	1PPS interrupt missed or GPS lock lost for >3 polling cycles	Elmoataz Abdellah & Seunghun Lee
21	Pass Sequencer - Pointing & Warmup Timing [L2-SYS-039–041, L4-OBC-234–243]	ADCS fine pointing delay or camera 60s warmup delay misses the Lake Erie imaging window	Missed 3-day revisit Lake Erie imaging passes (L1-SCI-014) or motion-blurred imagery (>90 m GSD)	4	Brief orbital imaging windows requiring tight sequencing of power rails, 60s camera warmup, and ADCS lock	3	ADCS 30-minute orbit propagation packets (0x43) and Ground Station pre-pass TLE validity filtering	19	-	Technical	Reduce	ADCS asserts MSG_ADCS_POINTING_TIMEOUT (0x44) or camera warmup exceeds 60s (MSG_IMG_HARDWARE_FAULT)	Elmoataz Abdellah & Seunghun Lee
22	Telecommand Security - Ascon-AEAD128 Authentication [L2-MIS-079, L4-OBC-185–190]	Unauthorized command injection or replay attack violates spacecraft telecommand integrity	Accidental or malicious disruption of spacecraft state or uncommanded orbital maneuvers	5	Shared UHF transceiver handling unencrypted public packets alongside privileged telecommands	2	Ground Station pre-transmission syntax validation (L3-GND-232) and UART frame checksums	17	-	Technical	Avoid	Failed Ascon-AEAD128 authentication tag or duplicate sequence counter (L4-OBC-187, L4-OBC-189)	Kristen Griessman
23	Critical Command Safeguards - Arm/Fire Confirmation [L3-OBC-129–132, L4-OBC-193]	Unconfirmed critical command execution (Reboot, Power Cycle, RF Inhibit, Purge Image)	Accidental disruption of spacecraft operations or premature deletion of stored science imagery	5	Operator command syntax error or corrupted single telecommand uplink frame	2	Ground Station pre-transmission syntax validation (L3-GND-232) and telecommand verification	17	-	Technical	Avoid	Single unconfirmed critical command received without valid ARM/FIRE token (0x05, 0x06, 0x07, 0x09, 0x14)	Kristen Griessman
24	Amateur Digipeater Isolation & Buffering [L4-OBC-206–208]	Amateur digipeater traffic floods command buffers and starves privileged ground telecommands	Loss of command link responsiveness and battery drain from continuous repeater packet handling	4	Shared UHF transceiver handling both public amateur radio digipeater packets (0x02) and telecommands	3	Software queue segregation and UART frame checksum validation	19	-	Technical	Avoid	Amateur digipeater ring buffer utilization >90% or unserviced telecommand packet drop	Kristen Griessman
25	Power Subsystem - Battery State of Charge [L2-SYS-048, L4-OBC-237]	Battery depletion (<15–20% SoC) during high-power imaging or high-rate S-Band downlink passes	Spacecraft brownout reset, ungraceful flash shutdown, or permanent battery cell capacity loss	5	High peak power draw of multispectral imagers, RP2350 processor, and S-Band PA during pass	3	60-second subsystem telemetry polling (L4-OBC-220) and EPS hardware overcurrent protection	21	-	Technical	Reduce	Battery SoC <20% pre-pass check (L4-OBC-237) or <15% low-battery alert (0x51)	Caden O'Leary & Seunghun Lee
26	Post-Deployment Inhibit Timers & Brownout [L3-SYS-037, L2-MIS-071, L4-OBC-210–216]	Brownout reset during 30-min deployer or 45-min RF post-deployment inhibit countdown timers	Premature deployable actuation or RF emission inside dispenser hazard zone violating launch safety	5	Power brownout during separation switch release or cold battery cell voltage dip upon dispenser exit	2	NRAM state persistence of elapsed seconds every 30s and EPS hardware separation switches	17	-	Technical	Reduce	Power brownout during startup timer countdown or unpersisted reboot during first 45 minutes	Caden O'Leary & Seunghun Lee
27	Thermal Monitoring & Automated Cutoffs [L4-OBC-221–222, L4-OBC-257]	Thermal excursion outside allowable limits (-40°C to 60°C internal; >45°C battery; >85°C S-Band PA)	Battery cell thermal degradation or S-Band power amplifier silicon failure	4	High localized heat dissipation in 3U structure during extended pass in direct solar soak	3	60-second subsystem temperature polling and EPS automated thermal rail shutdown	19	-	Technical	Reduce	Component temperature outside operating setpoints: Battery >45°C or S-Band PA >85°C (L4-OBC-257)	Caden O'Leary & Seunghun Lee
28	Flight Board Quarantine & Dev Board Testing [L1-PRJ-025]	EnduroSat flight boards quarantined from routine bench testing to prevent wear/damage	Engineering model testbench restricted to dev board/breadboard; subtle interface discrepancies missed	4	Strict $0 OBC budget freeze and single irreplaceable flight unit mandate quarantine from routine testing	3	Preservation protocol for flight board and near-identical MCU dev board with breadboards	19	-	External	Accept	Engineering model hardware validation blocked by dev board pinout differences or quarantine policy	Matthew Goldsberry & Caden O'Leary
29	Dev Board vs Flight Interface Rework	Dev board pinout or timing differences cause late rework during final flight unit acceptance	Cascading integration defects across interfaces discovered late, delaying flight acceptance sign-off	4	Subtle transceiver revisions and trace capacitance differences between dev board and flight OBC	3	Interface abstraction drivers (NASA F' OSAL) and formal breadboard hardware verification protocols	19	-	External	Accept	Bus timing jitter >10% nominal or transceiver logic level threshold mismatch on flight test	Matthew Goldsberry & Caden O'Leary
30	NASA CSLI Launch Delay & Orbit Parameters [L1-MIS-018-017, L1-PRJ-022–023]	CSLI launch delay or orbit parameter changes (40°–140° inc, 350–550 km alt) alter pass timings and eclipse	Hardcoded pass execution timings, thermal thresholds, or inhibit timers mismatched with assigned orbit	4	External dependence on NASA CSLI and commercial mission integrators (SpaceX, Nanoracks/Voyager, Tyvak)	3	Designing to worst-case CubeSat Design Specification Rev. 14.1 and 40°–140° orbital envelope	19	-	External	Accept	NASA CSLI manifest selection or Mission Integrator ICD specifies updated orbit parameters	Matthew Goldsberry
31	Federal Licensing (FCC/NOAA) & RF Compliance [L0-PRJ-004, L1-PRJ-018–021, L3-OBC-171]	Delayed FCC/IARU/NOAA licensing or uncommanded RF transmission violating 45-min post-deployment silence	Federal regulatory cease-transmission order, denial of launch clearance, or spectrum violation	5	Mandatory FCC, IARU, NOAA remote sensing, and US export control (ITAR/EAR) regulations	2	Coordination with UC Amateur Radio Club (UCARC) and UC Office of Research	17	-	External	Avoid	Regulatory compliance audit finding, FCC/IARU shutdown directive (Cmd 0x07), or unencrypted downlink	Kristen Griessman & Caden O'Leary
32	Internal CAN Bus Network - Watchdog & Heartbeat Supervisor [L4-OBC-250, L4-OBC-252]	CAN communication watchdog timeout or heartbeat supervisor thread lockup halts inter-subsystem health monitoring	Loss of subsystem health telemetry and failure to detect deadlocks across ADCS, EPS, or payload modules, delaying autonomous fault recovery	4	Blocking transmission calls, queue buffer saturation, or missed heartbeat intervals under heavy bus traffic	3	NASA F' Svc::Health component and periodic CAN heartbeat ping/pong protocol	19	-	Technical	Reduce	CAN communication thread blocked >500 ms or missing subsystem heartbeat response across 3 consecutive cycles	Matthew Goldsberry & Caden O'Leary

ID	Failure	Prev. Critical Number	Owner	Recommended Mitigation Action(s)	New Severity	New Occourance	New Critical Number	Reviewed	Approved	Category	Strategy	Trigger	Owner(s)
1	Radiation-induced Single-Event Upset (SEU) bit flips corrupt active flight software execution or dynamic RAM memory structures	24	OBC	Implement 3-WDT majority voting (Watchdog Architecture), FreeRTOS MPU stack guard regions, and autonomous thread recovery routines.	3	2	11	Yes	Yes	Technical	Reduce	Processor watchdog reset or unhandled RTOS HardFault exception	Seunghun Lee & Matthew Goldsberry
2	Single-Event Latch-up (SEL) or flash memory wear corrupts primary flight OS image in boot flash	21	OBC	Implement golden-image ROM bootloader fallback after 3 consecutive failed boots (L4-OBC-251) and 30s MRAM state persistence (L4-OBC-212).	3	1	5	Yes	Yes	Technical	Reduce	Bootloader CRC check failure on configuration/flight image or 3 consecutive failed boots (L4-OBC-251)	Seunghun Lee & Matthew Goldsberry
3	Shared CAN bus physical lockup or dominant-bit latch caused by a faulted peripheral transceiver	21	OBC	Implement MainComp_CANSupervisor to retry 3 times and sequentially command EPS to cut faulted subsystem power rails (L4-OBC 263).	3	1	5	Yes	Yes	Technical	Reduce	CAN controller Tx error counter exceeds 255 or bus dominant state persists >500 ms	Matthew Goldsberry
4	Transient EMI noise spikes induce repeated transmission frame errors, forcing CAN controller into bus-off state	19	OBC	Enforce automatic local CAN controller re-initialization sequence and transient error filtering before escalating to Safe Mode.	2	1	3	Yes	Yes	Technical	Reduce	CAN controller enters bus-off state across 3 consecutive local reset attempts (L4-OBC-252, L4-OBC-263)	Matthew Goldsberry
5	Schedule slip in Phase 1 Core Processing tasks cascades across subsequent tasks	23	OBC	Decouple sequential Table 1 dependencies using mocked hardware interfaces (L4-OBC-256) so Phase 2–4 tasks run concurrently; use Winter Break as buffer.	3	2	11	Yes	Yes	Team	Reduce	Any Phase 1 Core Processing task slips >48 hours past its scheduled deadline	Matthew Goldsberry
6	NASA F' framework autocoding learning curve and FreeRTOS integration complexity delay serial Table 1 tasks	22	OBC	Conduct targeted F' autocoder workshops, provide pre-built component templates, and pair senior developers with junior members.	2	2	8	Yes	Yes	Team	Reduce	NASA F' component integration or topology build blocked >72 hours on critical path	Matthew Goldsberry
7	Steep onboarding barrier for C++, FreeRTOS, and NASA F' environment setup deters club contributors	18	OBC	Containerize the NASA F' / FreeRTOS build environment using Docker/DevContainers for one-command setup across all student laptops.	2	1	3	Yes	Yes	Team	Avoid	New member dev environment setup takes >2 weeks to compile/run a first F' build	Caden O'Leary
8	Undocumented breadboard/UART jumper wiring changes between lab sessions cause accidental wiring faults	15	OBC	Maintain a formalized physical testbench wiring schematic and pinout diagram in OBC GitHub README; label all bench jumpers.	2	1	3	Yes	Yes	Team	Avoid	Lab testbench wiring modified without updating pinout schematic or wiring discrepancy found	Caden O'Leary
9	Breadboard jumper wire signal degradation, parasitic capacitance, or loose connections cause bus errors	18	OBC	Use rigid short-length jumpers, dedicated ground planes, and logic analyzer bus verification before running multi-hour HIL test scripts.	2	1	3	Yes	Yes	Technical	Avoid	Breadboard bus signal integrity failure or peripheral pin assignment mismatch with flight schematic	Elmoataz Abdellah
10	Clocking, memory, or peripheral register timing discrepancies between dev MCU and EnduroSat flight board delay bug detection	19	OBC	Enforce strict software-in-the-loop (SITL) unit tests achieving >=80% coverage (L4-OBC-256) and formalize hardware register abstraction layer.	2	1	3	Yes	Yes	Technical	Avoid	Off-target unit test coverage drops below 80% (L4-OBC-256) or flight board acceptance anomaly	Elmoataz Abdellah
11	Catastrophic hardware failure or unrecoverable lockup of Main Computer severs primary command and telemetry	17	OBC	Transfer emergency beaconing and command handling to EPS Power Computer via backup links to UHF and S-Band radios (L2-SYS-062, L3-POW-197).	3	1	5	Yes	Yes	Technical	Transfer	Main Computer fails to transmit periodic heartbeat to EPS within L4-POW-174 timeout after EPS power-cycle	Caden O'Leary & Seunghun Lee
12	Desynchronization or timing drift among the 3 concurrent WDTs triggers false-positive Safe Mode trip	19	OBC	Implement a time-deterministic WDT reset strobe with 2-of-3 majority voting (Watchdog Architecture) and synchronize tick windows.	2	1	3	Yes	Yes	Technical	Reduce	1 of 3 WDTs disagrees with the other 2 (2-of-3 majority mismatch)	Matthew Goldsberry & Caden O'Leary
13	Unmonitored communication hang or serial lockup on dynamically toggled UART interface lines	15	OBC	Add explicit UART timeout watchdog supervisors and state recovery handlers in NASA F' Svc::Health component (Peer Review Action #4).	2	1	3	Yes	Yes	Technical	Reduce	Dynamic UART state transition timeout or unserviced serial thread blocked >500 ms	Matthew Goldsberry & Caden O'Leary
14	Runtime SRAM memory utilization exceeds 95% capacity under concurrent workload	19	OBC	Allocate all static memory buffers at boot (Task 001, L4-OBC-248) and suspend non-essential logging tasks if RAM exceeds 95% (L4-OBC-249).	2	1	3	Yes	Yes	Technical	Reduce	Runtime RAM utilization exceeds 95% threshold alert (L4-OBC-249)	Elmoataz Abdellah & Seunghun Lee
15	Flash mass storage saturation (>99%) or bad NAND flash blocks corrupt stored Lake Erie imagery	19	OBC	Remap bad flash blocks (L4-OBC-227) and overwrite only oldest unflagged telemetry logs (FIFO) at 99% while preserving unpurged science frames (L4-OBC-230).	2	1	3	Yes	Yes	Technical	Reduce	Flash storage utilization >85% warning (L4-OBC-229), >99% capacity (L4-OBC-230), or SPI flash CRC error	Elmoataz Abdellah & Seunghun Lee
16	SPI bus contention between high-rate Imager writes and Telemetry logging threads	15	OBC	Serialize SPI storage access via FreeRTOS mutex locks, ring buffering, and DMA priority channels (L4-OBC-253).	2	1	8	Yes	Yes	Technical	Reduce	SPI transaction lock timeout or bus collision flag asserted during active pass	Elmoataz Abdellah & Seunghun Lee
17	Data Processor arithmetic divide-by-zero or overflow during Cyanobacteria/Chlorophyll Index calculation	16	OBC	Implement Processor_MathGuard bounds-checking to trap div-by-zero and fall back to lossless-compressed raw image storage (L4-OBC-223, L4-OBC-246).	2	1	8	Yes	Yes	Technical	Reduce	Arithmetic divide-by-zero or numeric overflow exception trap (L4-OBC-246)	Matthew Goldsberry & Seunghun Lee
18	Data Processor hangs or exceeds execution timeout budget during Lake Erie image processing	16	OBC	Enforce a hard execution timer (CMD_DP_ABORT, L4-OBC-262), power-gate processor during standby (L4-OBC-261), and downlink raw thumbnails via UHF (0x13).	2	1	8	Yes	Yes	Technical	Reduce	Data Processor execution timer expiry (L4-OBC-262) or task watchdog timeout	Matthew Goldsberry & Seunghun Lee
19	Main Computer and Data Processor contend for S-Band radio access causing buffer underrun/stall	18	OBC	Implement strict time-division multiplexing for S-Band radio access and up to 3 UART frame block retries (L4-OBC-245).	2	1	8	Yes	Yes	Technical	Reduce	S-Band transmitter buffer underrun or overflow flag asserted (L4-OBC-200)	Matthew Goldsberry & Seunghun Lee
20	GPS lock or 1PPS signal dropout during orbital eclipse causes epoch clock timing drift	18	OBC	Use monotonic <=20 ppm tick counter rather than wall-clock subtraction during eclipse (L4-OBC-217, L4-OBC-243) and ground sync (0x16).	2	2	8	Yes	Yes	Technical	Reduce	1PPS interrupt missed or GPS lock lost for >3 polling cycles	Elmoataz Abdellah & Seunghun Lee
21	ADCS fine pointing delay or camera 60s warmup delay misses the Lake Erie imaging window	19	OBC	Gate shutter trigger on both MSG_ADCS_TARGET_ACQUIRED and MSG_IMG_WARMUP_READY (L4-OBC-239) and abort cleanly to Standby on timeout (L4-OBC-241).	2	2	8	Yes	Yes	Technical	Reduce	ADCS asserts MSG_ADCS_POINTING_TIMEOUT (0x44) or camera warmup exceeds 60s (MSG_IMG_HARDWARE_FAULT)	Elmoataz Abdellah & Seunghun Lee
22	Unauthorized command injection or replay attack violates spacecraft telecommand integrity	17	OBC	Authenticate and decrypt all non-amateur telecommands with Ascon-AEAD128 and anti-replay sequence counters (L4-OBC-185–189).	3	1	6	Yes	Yes	Technical	Avoid	Failed Ascon-AEAD128 authentication tag or duplicate sequence counter (L4-OBC-187, L4-OBC-189)	Kristen Griessman
23	Unconfirmed critical command execution [Reboot, Power Cycle, RF Inhibit, Purge Image]	17	OBC	Require strict two-step ARM / FIRE sequence within 30-second window for all critical telecommands (L4-OBC-193).	3	1	6	Yes	Yes	Technical	Avoid	Single unconfirmed critical command received without valid ARM/FIRE token (0x05, 0x06, 0x07, 0x09, 0x14)	Kristen Griessman
24	Amateur digipeater traffic floods command buffers and starves privileged ground telecommands	18	OBC	Isolate amateur digipeater traffic in a rate-limited 64 kb ring buffer with strict priority preemption for telecommands (L4-OBC-207–208).	2	1	8	Yes	Yes	Technical	Avoid	Amateur digipeater ring buffer utilization >90% or unserviced telecommand packet drop	Kristen Griessman
25	Battery depletion (<15–20% SoC) during high-power imaging or high-rate S-Band downlink passes	21	OBC	Preemptively process low-battery alerts within 100 ms (L4-OBC-222) to abort pass, gracefully shut down subsystem rails with 50 ms staggered switching (L4-OBC-211).	3	1	6	Yes	Yes	Technical	Reduce	Battery SoC <20% pre-pass check (L4-OBC-237) or <15% low-battery alert (0x51)	Caden O'Leary & Seunghun Lee
26	Brownout reset during 30-min deployer or 45-min RF post-deployment inhibit countdown timers	17	OBC	Persist cumulative 30-min/45-min inhibit timer seconds and mode flags in non-volatile MRAM every 30s across resets (L4-OBC-212, L4-OBC-213).	3	1	6	Yes	Yes	Technical	Reduce	Power brownout during startup timer countdown or unpersisted reboot during first 45 minutes	Caden O'Leary & Seunghun Lee
27	Thermal excursion outside allowable limits (-40°C to 60°C internal, +45°C battery, +85°C S-Band PA)	19	OBC	Implement automated thermal cutoffs in flight software (L4-OBC-257) and power-cycle/throttle RF amplifier during elevated temperature soak.	2	1	8	Yes	Yes	Technical	Reduce	Component temperature outside operating setpoints: Battery >45°C or S-Band PA >85°C (L4-OBC-257)	Caden O'Leary & Seunghun Lee
28	EnduroSat flight boards quarantined from routine bench testing to prevent wear/damage	19	OBC	Enforce strict quarantine protocol, use near-identical MCU dev board with breadboards for EM, and utilize NASA F' hardware abstraction layer.	2	2	8	Yes	Yes	External	Accept	Engineering model hardware validation blocked by dev board pinout differences or quarantine policy	Matthew Goldsberry & Caden O'Leary
29	Dev board pinout or timing differences cause late rework during final flight unit acceptance	19	OBC	Implement benchtop power limiting, interface abstraction drivers (NASA F' OSAL), and formal breadboard hardware verification protocols.	2	2	8	Yes	Yes	External	Accept	Bus timing jitter >10% nominal or transceiver logic level threshold mismatch on flight test	Matthew Goldsberry & Caden O'Leary
30	CSLI launch delay or orbit parameter changes (401–140° inc, 350–550 km alt) alter pass timings and eclipse	19	OBC	Store all operational thresholds, slew lead times, and inhibit timers in a telecommand-reconfigurable non-volatile parameter table (L3-OBC-175, L4-OBC-264).	2	2	8	Yes	Yes	External	Accept	NASA CSLI manifest selection or Mission Integrator ICD specifies updated orbit parameters	Matthew Goldsberry
31	Delayed FCC/IARU/NOAA licensing or uncommanded RF transmission violating 45-min post-deployment silence	17	OBC	Enforce hardware-persisted 45-minute RF inhibit timers (L4-OBC-213), immediate ground RF inhibit commands (0x070x08), and payload encryption.	2	1	8	Yes	Yes	External	Avoid	Regulatory compliance audit finding, FCC/IARU shutdown directive (Cmd 0x07), or unencrypted downlink	Kristen Griessman & Caden O'Leary
32	CAN communication watchdog timeout or heartbeat supervisor thread lockup halts inter-subsystem health monitoring	19	OBC	Implement non-blocking FreeRTOS CAN message queues with bounded timeout handling and an active heartbeat monitor thread to detect and reset stalled transceiver drivers.	2	1	8	Yes	Yes	Technical	Reduce	CAN communication thread blocked >500 ms or missing subsystem heartbeat response across 3 consecutive cycles	Matthew Goldsberry & Caden O'Leary